Individual Contributions Report
Character Controller (Student 3)

Table 1: Student details

Name	N R Yenash
Student ID	w2181809
Group / Role	Group 02, Student 3
Module	5MMCS007C 3D Interactive Media, CW1
GitHub folder	https://github.com/iSaTHsiNDuMe/rabbit-carrot-collector-cw1/tree/Student-3/Group02_CW1_Risal_Yenash_w2181809

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1. Introduction and role

Our group build *Rabbit Survivl*, a LEGO-style Unity 6 scene using the Universal Render Pipelin. The player controls a rabbit that collects carrots, avoids two hunters by hiding in bushe, crosses a bridge and meets a second rabbit in a cave. The scene combines each members tutorial brick into one playable diorama.

As Student 3, my brief was the Character Controller brick from Tutorials 02 and 03, plus a LEGO minifig with a walk animation controlled by a Unity UI button from Tutorial 05.2. I went beyond the minimum by turning the controller into the player's rabbit. I added camera-relative movement, jumping, hiding in bushes with on-screen feedback, a third-person camera that avoids obstacles, and mouse camera controls.

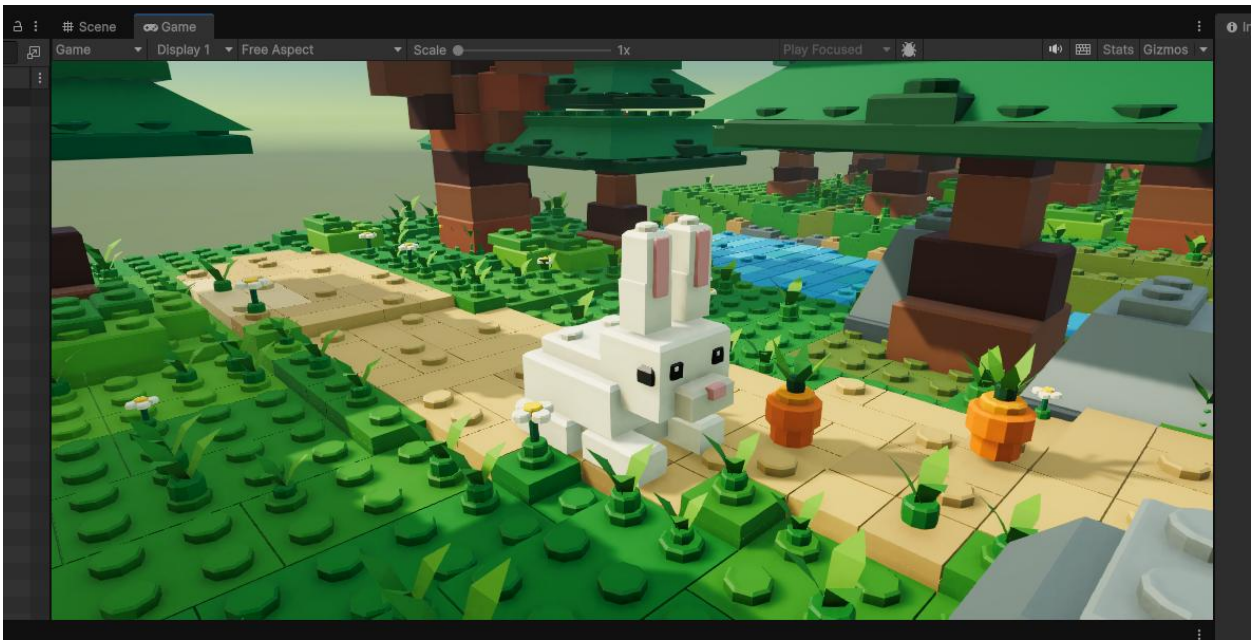


Figure 1: Wide shot of the RabbitSurvivalLEGO scene with the rabbit visible

2. Character Controller brick (tutorial work)

I first built brick in my own Unity project, Member3_CharacterController, before bringing the ideas into the group scene. A LEGO brick called PlayerBrick carry Unity's CharacterController component and my BrickController script.

A Character Controller differ from Student 1's Rigidbody brick. A Rigidbody is moved by the physic engine through forces and mass. A Character Controller is moved directly by script with Move(), but still collides with the world, slides along walls and climbs small steps. This give precise, responsive control, which suits a player character.

Each frame the script reads the horizontal and vertical input axes, builds a movement vector and passes it to CharacterController.Move. Gravity is handled manually: a vertical velocity is reduced every frame and reset when the brick is grounded. Pressing Space while grounded set upward velocity to make the brick jump. Building it taught me how frame rate independent movement works using Time.deltaTime, and why gravity must be added by hand when there no Rigidbody.

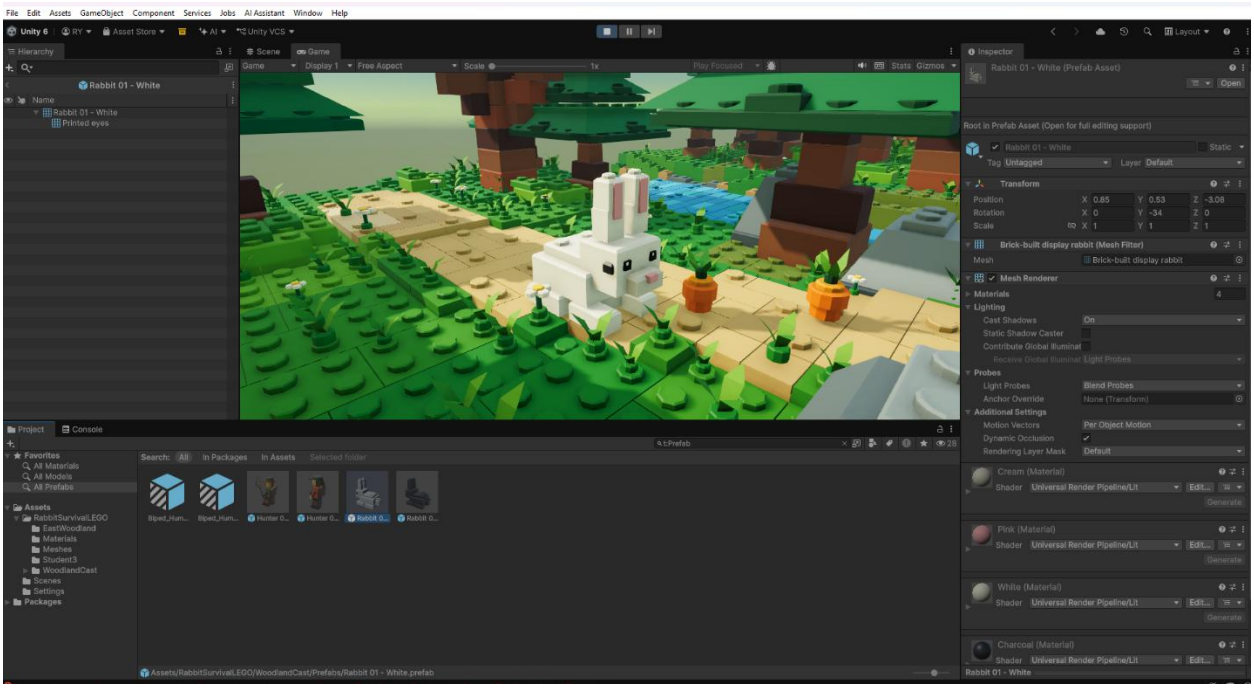


Figure 2: Rabbit 01 - White prefab in Game view, with its Inspector

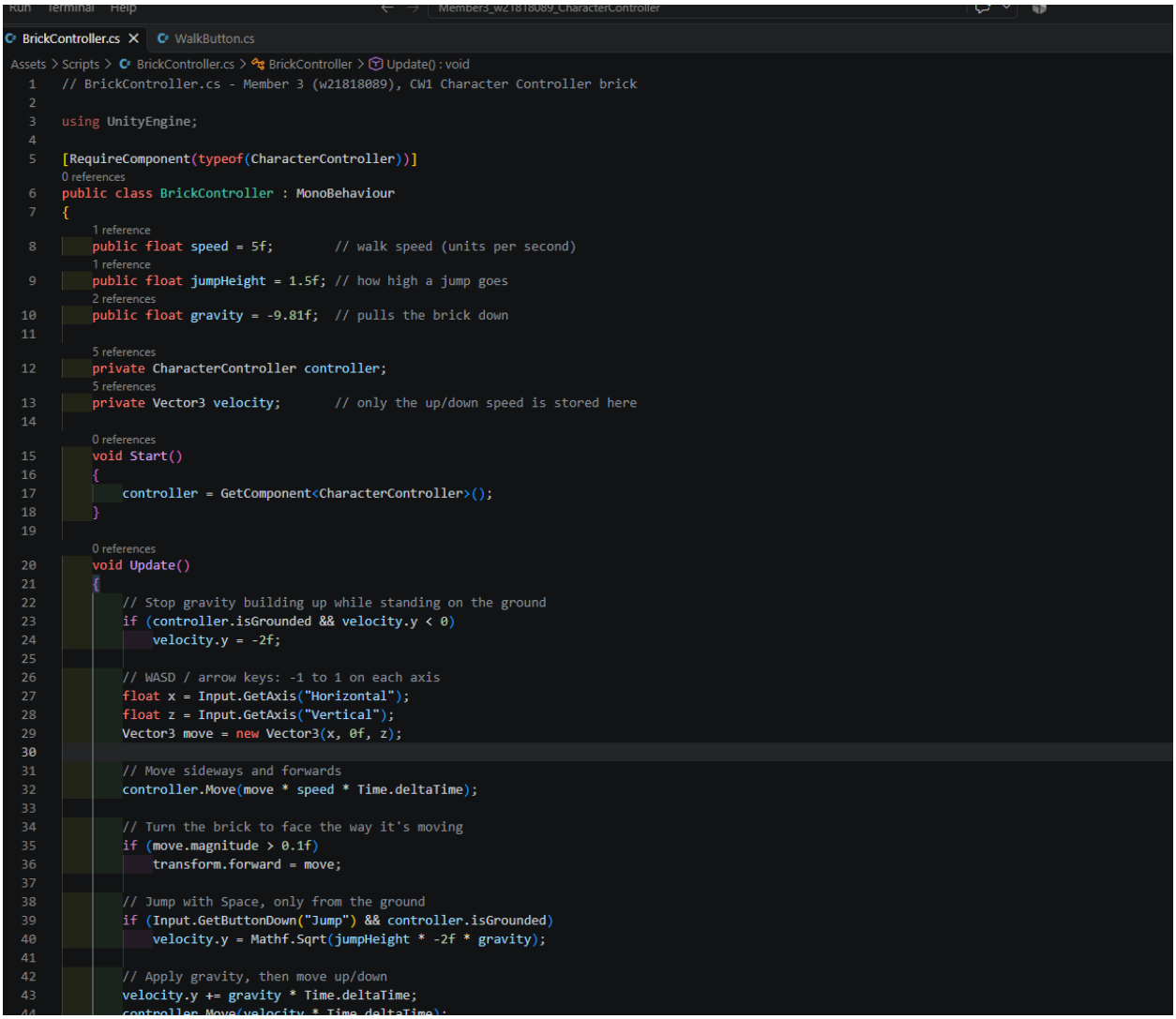


Figure 3: BrickController.cs in VS Code

3. LEGO minifig walk and UI button

For the minifig requiremnt I used the provided Lego Walk model and animated it with my Student3WalkPreview script. A UI panel with a Walk button appear only when the rabbit is within five unit of the minifig. This keeps the screen clear and ties the interface to the player's position in the worldd. The distance test uses sqrMagnitude against 25, which avoids a square rot every frame.

The button's OnClick event calls PlayWalk(), which starts a 2.4-second timer. In Awake() the script stores each limb's starting rotation so the animation alway returns to the rest pos. During the walk, a sine wave swings the limbs up to 29 degres. Even numbered limbs swing forward and odd ones backward, giving the opposite arm and leg motion of a real walk. The swing fades out over the last ffth of a second so it doesnt stop abruptly. An earlier version in my own project used a simple WalkButton script with a ToggleWalk function.

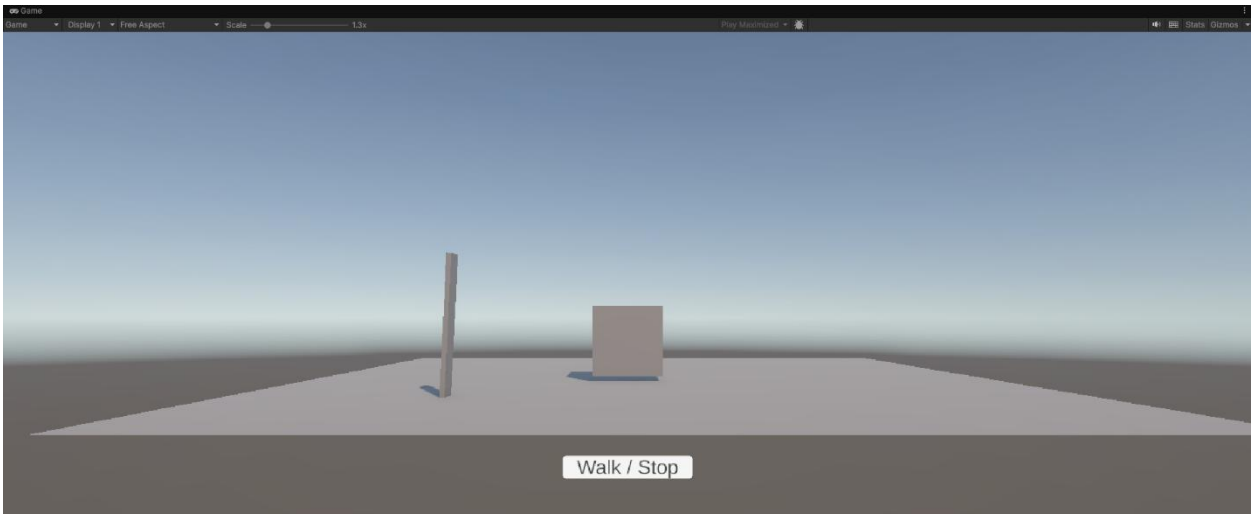


Figure 4: Minifig with the Walk / Stop button in Game view



Figure 5: Walk minifig Inspector showing the Animator and Walk Button script

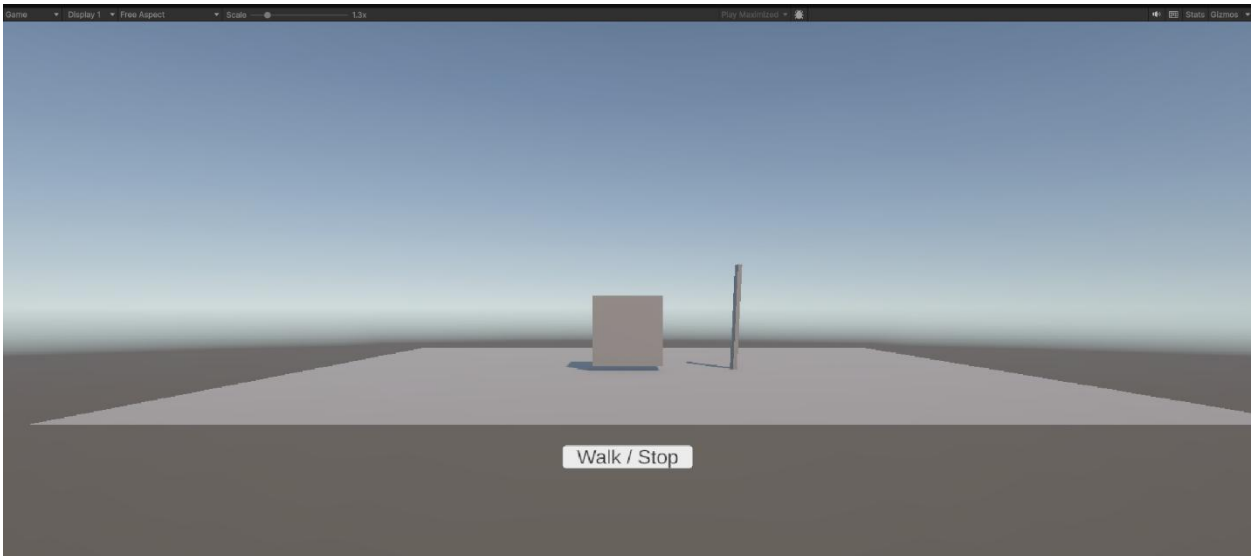


Figure 6: Minifig mid-walk

4. Rabbit controller in the group scene

In the group project the character controller became Student3RabbitController, atached to the rabit model. It uses Unity's new Input System, which the group project is set to, and RequireComponent guarantees a Character Controler is always present.

Camera relative movement. W, A, S and D build a direction that is clamped so digonal movement isnt faster. The direction is then rotated by the camera yaw, so W always moves away from the camera. The rabbit turns to face its travel direction with RotateTowards at 720 degrees per second, which is quick but smooth. In 3DUI terms this is a steering travel technique, and it match what players expect from third person game.

Jumping and gravity. When grounded, Space gives an upward speed of 5.25. Gravity of 17 units/s² is applied each frame with a capped fal speed. Movement and gravity go into a single controller.Move call per frame, which keeps colision and the grounded check reliable.

Hiding in bushes. The scene has six bush zones. Each frame, UpdateCover converts the rabbits position into each bush's local space and checkss whether it is inside the box. If so, IsHidden becomes true and a "Hidden" badge appear on a screen space canvas. Other scripts, such as the hunters, can read IsHidden, which links my work to the group's stealth gameplay. The badge gives the player immediate feedback, a key 3DUI principle.

Third person follow camera. The camera orbits a point just above the rabit. A SphereCast from the rabbit towards the camera detects trees and walls and puls the camera in front of them, so the rabbitt never disappear from view. The position is smothed with Lerp in LateUpdate, after the rabbit has moved, to avoid jitter.

Mouse controls. Pressing C locks the cursor for mouse look and Esc releases it. I later added holding the right mouse button to orbit and the scroll wheel to zoom between 2 and 9 units. Camera pitch is clamped between 10 and 65 degres so it cant go under the ground.

Table 2: Key rabbit controller settings

Feature	Setting	Purpose
Turn speed	720°/s (RotateTowards)	Quick but smooth facing of travel direction
Jump speed	5.25 upward	Jump from the ground with Space
Gravity	17 units/s², capped fall speed	Manual gravity without a Rigidbody
Hiding	6 bush zones (local-space box test)	Sets IsHidden and shows the Hidden badge

Feature	Setting	Purpose
Camera collision	SphereCast, Lerp in LateUpdate	Keeps rabbit in view without jitter
Zoom	2–9 units (scroll wheel)	Adjustable camera distance
Camera pitch	Clamped 10°–65°	Stops the camera going under the ground
Cursor	C locks / Esc releases; right mouse orbits	Mouse look controls

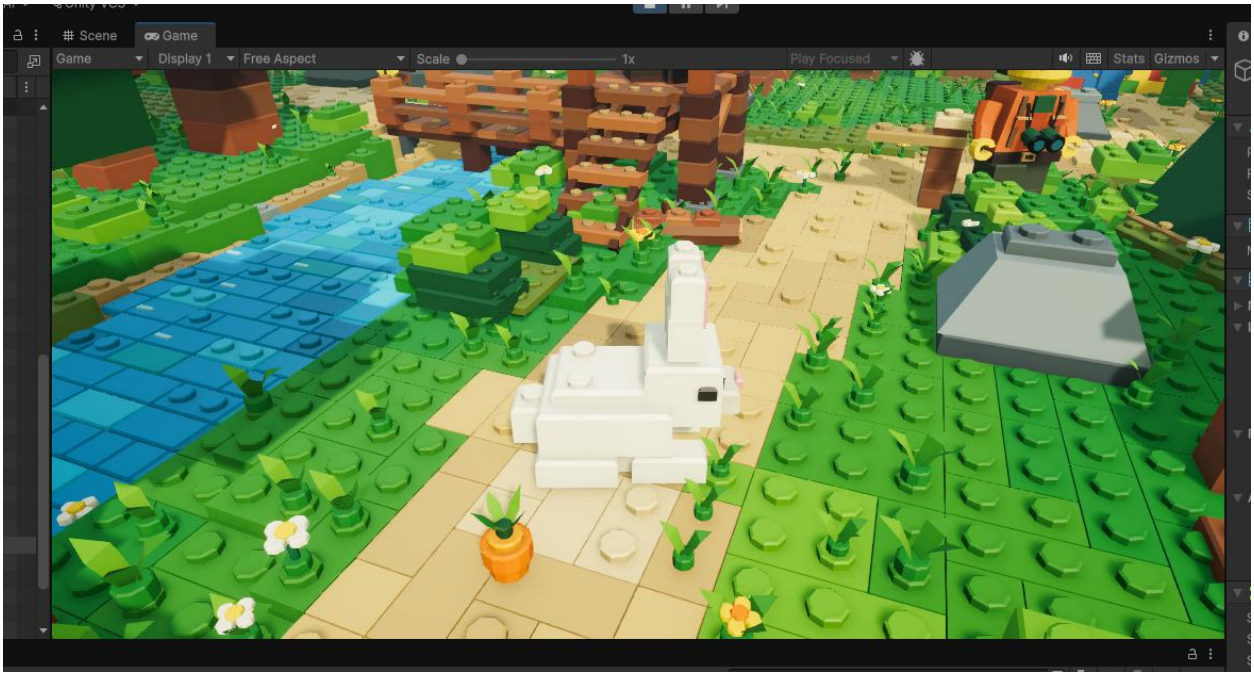


Figure 7: Rabbit moving along the trail in Game view

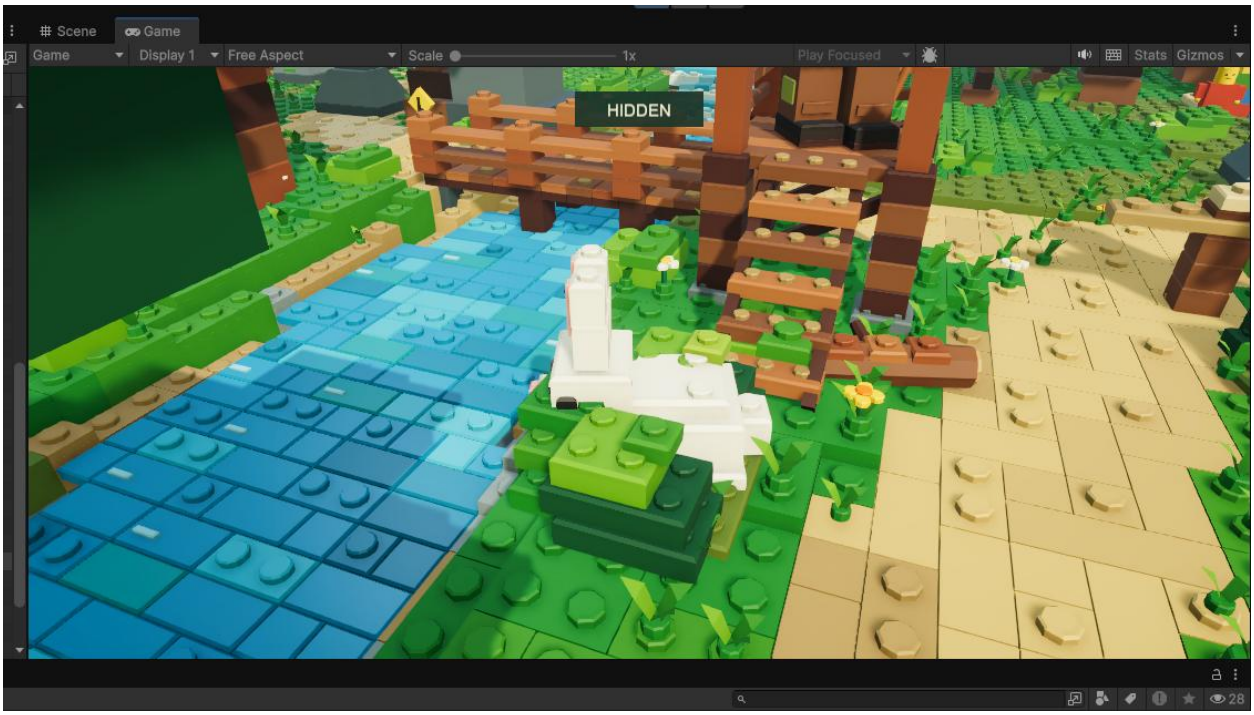


Figure 8: Rabbit inside a bush with the Hidden badge showing

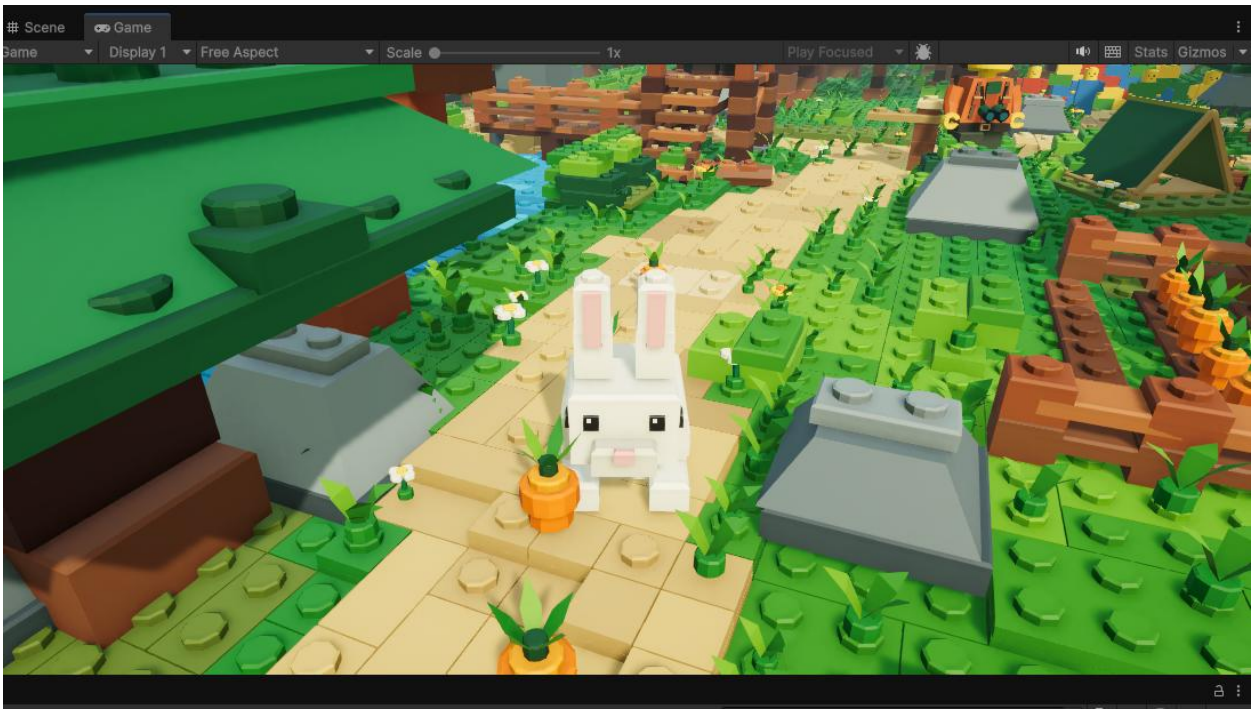


Figure 9: Camera pulled close by a tree

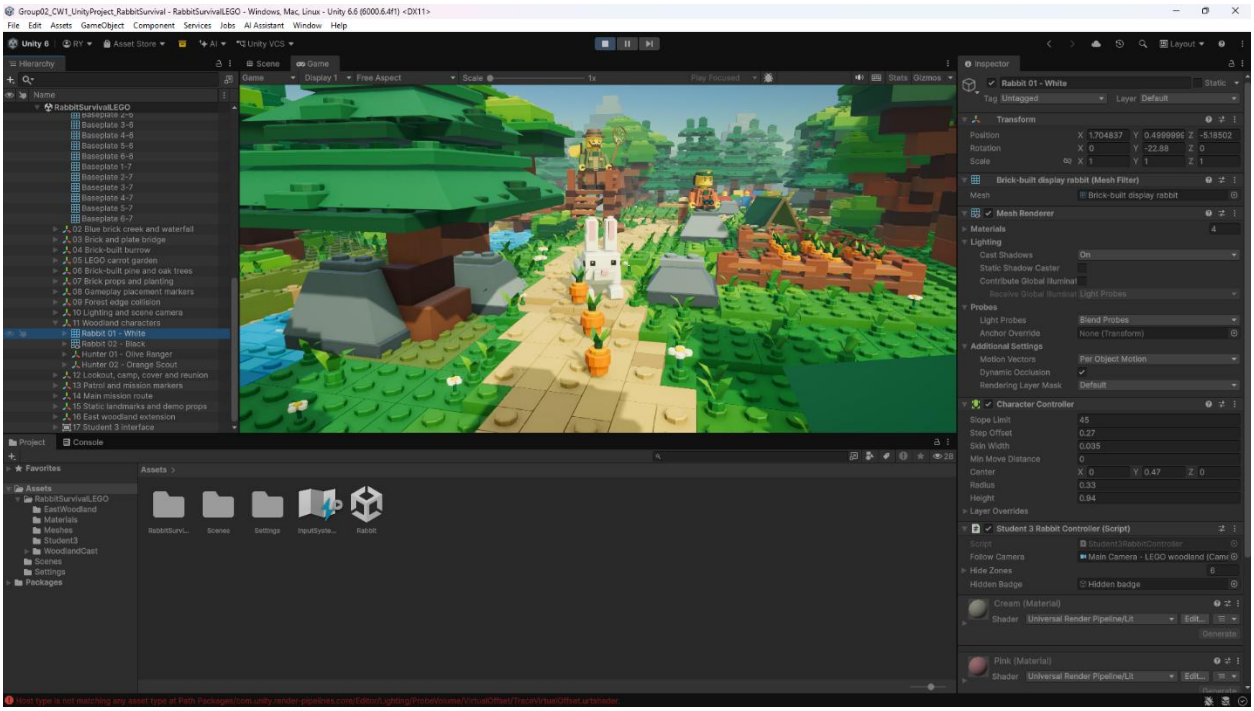


Figure 10: Inspector for Rabbit 01 - White showing Character Controller and Student3RabbitController

5. Integration and versioning

The group keep all work in one GitHub repository, rabbit-carrot-collector-cw1, containing the Unity project Group02_CW1_UnityProject_RabbitSurvival. My scripts in Assets/RabbitSurvivalLEGO/Student3. I used GitHub Desktop to manage my changes. To avoid overwriting other members' work on main, I created my own branch, Student-3, and pushed two commits to it. The first contained the rabbit controls, hiding, follow camera and walk preview, with the scene. The second added the mouse control. The branch can be merged into main through a pull request once the group has reviewed it.

Table 3: Student-3 branch commits

Commit	Contents
1	Rabbit controls, hiding, follow camera and walk preview, with the scene
2	Mouse controls (right-click orbit, scroll zoom)

One integration isue was Unity versions. The group project uses 6000.6.4f1, while my own project was made in 6000.6.7f1. Opening the project in a newer editor upgrades it, so I work in the group's version to keep everyone compatible.

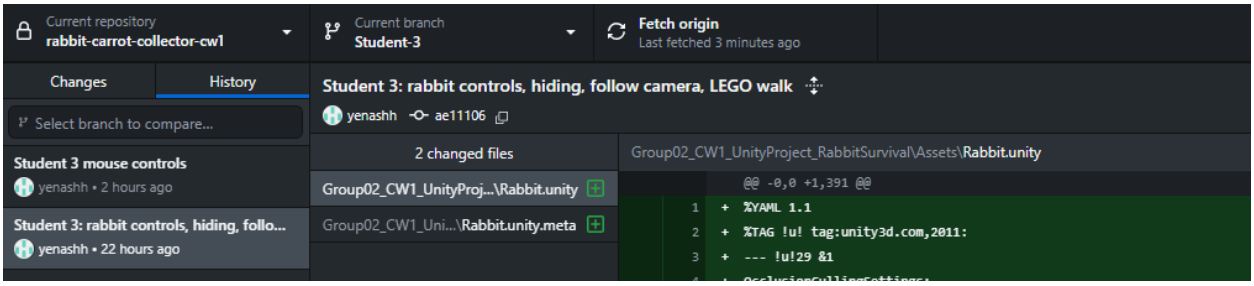


Figure 11: GitHub Desktop History showing the Student-3 commits

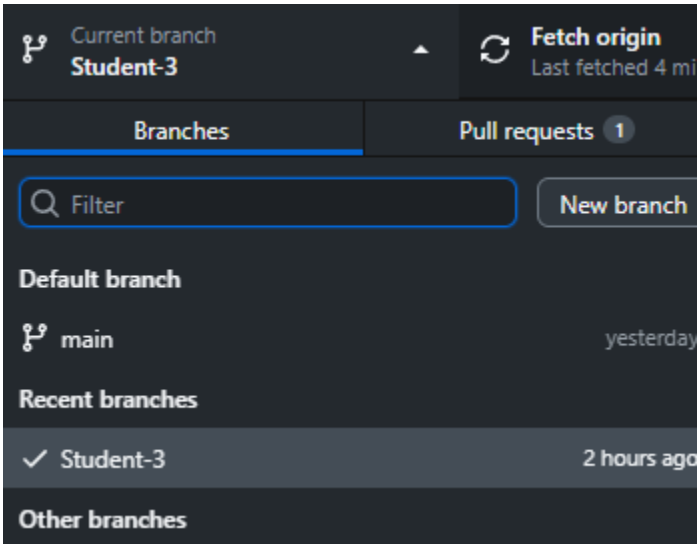


Figure 12: Student-3 branch in GitHub Desktop

6. Problems and debugging

The main problems I met, their causes and fixes are summarised in Table 4.

Table 4: Problems, causes and fixes

Problem	Cause	Fix / lesson
Jump not working the rabbit rarely jump	controller.Move called twice per frame (walk and gravity), making isGrounded unreliable	Combined both into one Move call per frame
Rabbit not moving in Play mode: keys did nothing, no errors	Unity's Pause button was on, so the game started paused	Turned Pause off learned to check the editor state before assuming the code is wrong
Camera clipping: camera went inside trees and wals	Fix camera distance	Sphere cast moves the camera closer when something is in the way

7. Conclusion and CW2 plans

My contribution gives the group responsive, playable rabbit with jumping, hiding, a camera that never loses the player, and a minifig walk trigered by a world aware UI button. Im most pleased with the hiding system because it conects my controller to the group's stealth gameplay.

For CW2 I plan to take the roles **Coding: Movement and camera** and **Coding: Interaction**. My next steps are to make the hunter react to IsHidden, give the rabbit a proper animated hop, and split the controler into smaller, reusable scripts.

Links:

GitHub individual folder: https://github.com/iSaTHsiNDuMe/rabbit-carrot-collector-cw1/tree/Student-3/Group02_CW1_Risal_Yenash_w2181809

Student 3 branch: https://github.com/iSaTHsiNDuMe/rabbit-carrot-collector-cw1/tree/Student-3

AI generated summary of AI chat transcripts

The student used Claude to suport their Student 3 work on the Chaacter Controller. Claude first gave fedback on the group’s WIP slides and drafted a WIP voiceover.

For the tutorial brick, Claude gave step by step guidance and the student createdd the Unity project and scene, and set up the PlayerBrick and its Character Controler themselves. Claude then gave a line by line plan and provided the BrickController script.

In the group project the student decided which features the rabbit neded and asked for each improvement, such as hiding in bushes, a follow camera and mouse controls. Claude provided the scripts for rabbit movement, hiding, the follow camera and carots, and tested them in Unity with the studentt. Together they fix a jump bug caused by two Move calls per frame. The student play tested carrot colecting and collected all 20 car ots. When the rabbit later stopped moving, the student noticed and reported it, and Claude found that Unity was paused.

the student recorded their own video and voiceover. For versioning, Claude explained GitHub Desktop, set up a Student-3 branch and aded right click look and scroll zoom the student organised their individual folder on GitHub.